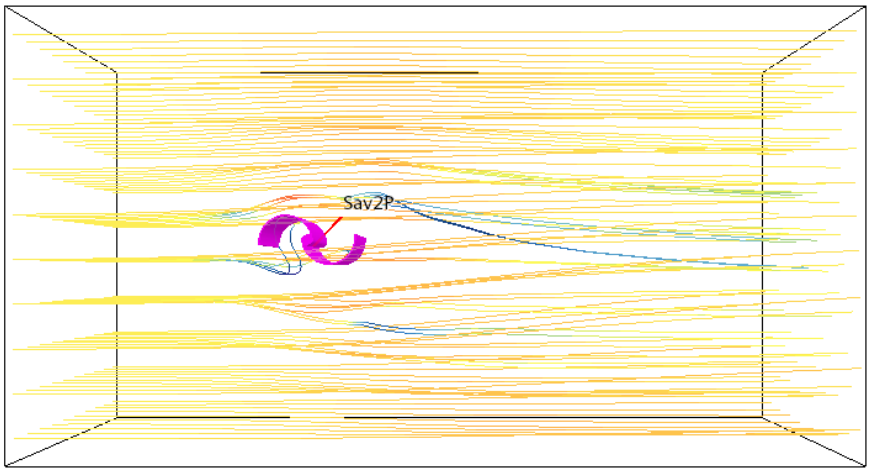
Institut Polytechnique des Sciences Avancées (IPSA)

Fluids Dynamics – Mf421



Laboratory Report

TP CFD Star CCM+ N°3

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Introduction

Savonius wind turbines are vertical-axis turbines that convert wind energy into rotational motion. Compared to horizontal-axis turbines, they are simpler in design and can operate regardless of wind direction. They are also well suited for low-speed and turbulent flow conditions, which makes them relevant for small-scale or urban applications.

In this tutorial, a two-bladed Savonius turbine is studied using computational fluid dynamics (CFD) with *Star-CCM+*. The objective is to analyze the flow around the rotor and evaluate its aerodynamic performance. In particular, quantities such as the power coefficient and the moment coefficient are investigated.

The simulation also makes it possible to observe flow features such as wake development and unsteady effects linked to vortex shedding. In addition, the influence of parameters like the Reynolds number and the tip speed ratio is examined in order to better understand the turbine behavior.

Analytical Part

The objective of this part is to determine the initial freestream velocity, the rotational velocity of the rotor, and the appropriate time step for the numerical simulation. The analysis covers two flow cases at different Reynolds numbers.

Case 1: Re = 12000

Freestream Velocity Calculation

The freestream velocity is determined using the Reynolds number definition:

$$Re = \frac{\rho \cdot d \cdot U_\infty}{\mu} \tag{1}$$

Solving for the freestream velocity U_∞:

$$U_\infty = \frac{\mu \cdot Re}{\rho \cdot d} = \frac{1.86 \times 10^{-5} \times 12000}{1.2 \times 0.1} = 1.86 \text{ m/s} \tag{2}$$

where $\mu = 1.86 \times 10^{-5}$ Pa·s is the dynamic viscosity of air, $\rho = 1.2$ kg/m^3 is the air density, and $d = 0.1$ m is the characteristic dimension.

Rotational Velocity Calculation

The rotational velocity is determined based on the tip-speed ratio (or λ parameter), which represents the ratio of rotor tip velocity to freestream velocity:

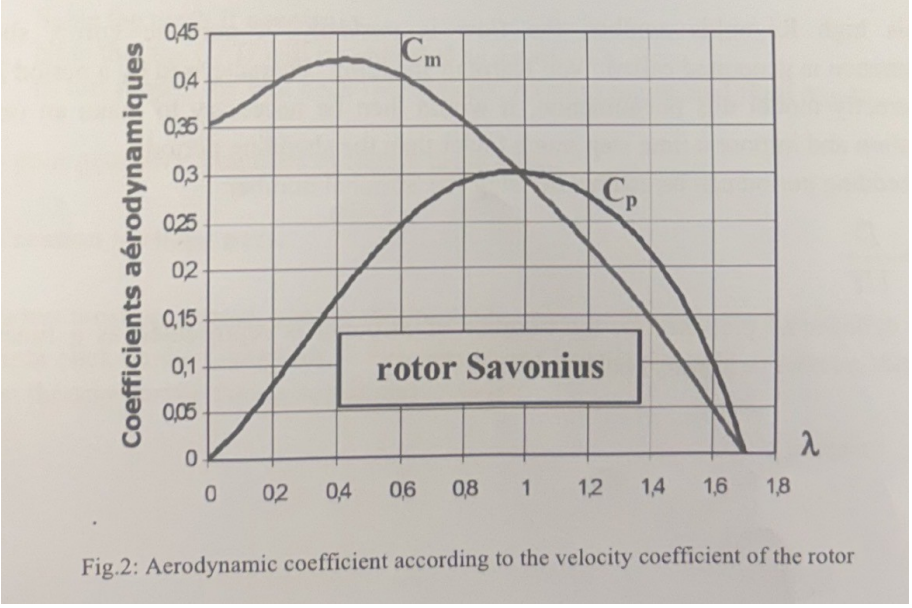


Figure 1: Power coefficient as a function of tip-speed ratio

From the power coefficient graph, we select $\lambda = 1$ to maximize the power delivered by the rotor. The angular velocity is then calculated as:

$$\lambda = \frac{\omega \cdot R}{U_\infty} \implies \omega = \frac{\lambda \cdot U_\infty}{R} = \frac{1 \times 1.86}{0.1} = 18.6 \text{ rad/s} \tag{3}$$

where $R = 0.1$ m is the rotor radius.

Time Step Selection

To capture the rotor dynamics accurately, the time step is determined by discretizing one complete rotation into angular increments of 10. First, the period of rotation is calculated:

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{18.6} = 0.338 \text{ s} \tag{4}$$

The time step corresponding to a 10 angular increment is:

$$\Delta t = \frac{T}{36} = \frac{0.338}{36} = 9.4 \times 10^{-3} \text{ s} \tag{5}$$

This time step ensures sufficient temporal resolution to capture the aerodynamic phenomena associated with rotor rotation while maintaining computational efficiency.

Case 2: Re = 4000

Freestream Velocity Calculation

The freestream velocity is determined using the Reynolds number definition:

$$Re = \frac{\rho \cdot d \cdot U_\infty}{\mu} \tag{6}$$

Solving for the freestream velocity U_∞:

$$U_\infty = \frac{\mu \cdot Re}{\rho \cdot d} = \frac{1.86 \times 10^{-5} \times 4000}{1.2 \times 0.1} = 0.62 \text{ m/s} \tag{7}$$

where $\mu = 1.86 \times 10^{-5}$ Pa·s is the dynamic viscosity of air, $\rho = 1.2$ kg/m^3 is the air density, and $d = 0.1$ m is the characteristic dimension.

Rotational Velocity Calculation

From the power coefficient graph, we select $\lambda = 1$ to maximize the power delivered by the rotor. The angular velocity is then calculated as:

$$\lambda = \frac{\omega \cdot R}{U_\infty} \implies \omega = \frac{\lambda \cdot U_\infty}{R} = \frac{1 \times 0.62}{0.1} = 6.2 \text{ rad/s} \tag{8}$$

where $R = 0.1$ m is the rotor radius.

1. Geometry - Mesh

Geometry

The geometry of the Savonius turbine was already provided at the beginning of the study. Therefore, the work mainly consisted in building the computational domain around the rotor in order to correctly simulate the flow.



Figure 2: 3D Savonius Rotor

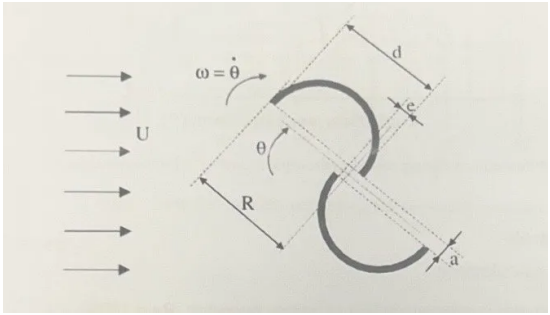


Figure 3: 2D Savonius Rotor

The geometry of the Savonius turbine was directly imported into *Star-CCM+*. The main objective of this step was therefore to define a computational domain suitable for the CFD simulation.

To achieve this, the fluid domain was divided into two distinct regions. The first one corresponds to a large outer domain, designed to represent the surrounding airflow and to ensure that inlet and outlet boundary conditions do not influence the turbine behavior. The second region is a cylindrical zone surrounding the turbine, which is used to model the rotational motion of the rotor.

This separation into two regions is essential for the simulation setup, as it allows the definition of a rotating reference frame in the vicinity of the turbine while keeping a stationary domain for the external flow. An interface is then created between the rotating and stationary regions to ensure proper coupling of the flow.

Finally, all boundaries were identified and assigned according to their physical role (inlet, outlet, symmetry planes, and interfaces), preparing the model for the meshing and physics definition steps.

Meshing

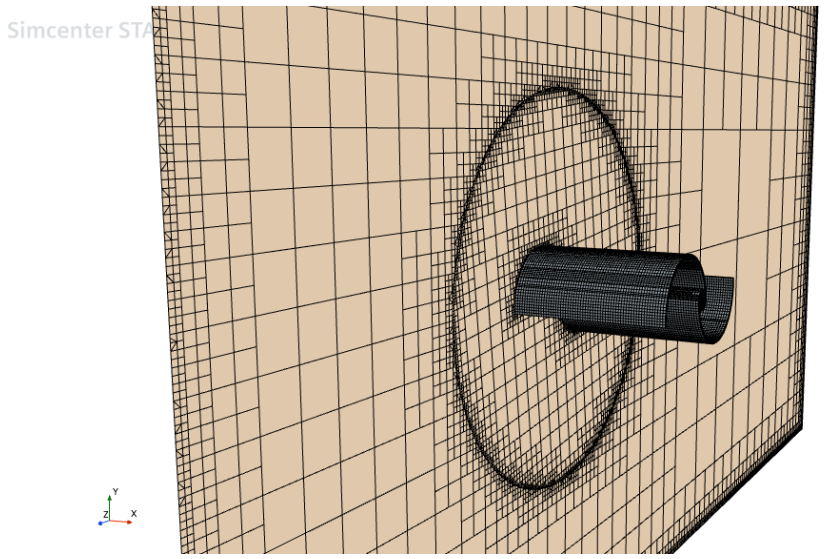


Figure 4: Vue globale du maillage (Mesh Scene)

The mesh was generated using the *Trimmer* meshing technique available in *Star-CCM+*, which provides a good compromise between mesh quality and computational cost for complex geometries. As indicated in the tutorial, two different mesh continua were defined: one for the rotating cylindrical region and one for the outer fluid domain.

The mesh size was controlled through a base size parameter, adjusted according to the characteristic dimensions of the problem. A finer mesh was applied in the rotating region (*Mesh Rot*) with a characteristic cell size of **[0.005m]** in order to accurately capture the strong velocity gradients and unsteady flow structures near the turbine blades. In contrast, a coarser mesh (*Mesh Flu*) with a characteristic cell size of **[0.5m]** was used in the outer domain to reduce the overall computational cost while still capturing the global flow behavior.

Surface refinement parameters were also adjusted to improve resolution near the turbine geometry. In particular, the relative minimum and target surface sizes were reduced to better capture the blade curvature and the flow separation.

Once the surface mesh was validated, the volume mesh was generated. A cross-sectional plane was used to visually inspect the mesh distribution and ensure a proper transition between refined and coarse regions.

Finally, a mesh quality assessment was performed using the skewness angle criterion. A histogram of skewness angles was analyzed to verify that no cells exceeded critical values (typically below 90°), ensuring numerical stability and reliability of the simulation results.

2. Physical model Justification

The simulation was performed under the assumption of an incompressible, isothermal flow with constant density, which is appropriate for low-speed wind turbine applications.

A segregated flow model was chosen as it offers a good compromise between accuracy and CPU cost for such a problem. The k-ω turbulence model with turbulence suppression was chosen to account for turbulence effects, as it is known to be efficient for simulating near-wall effects and separated flows, as seen in Savonius turbines.

Since the Reynolds number is relatively high (Re $\approx$ 12000), the flow is expected to be unsteady.

A rotational motion was applied to the cylindrical region where the turbine was located, and this allowed the blades to turn at a given angular velocity. This method ensures that the interaction between the rotating blades and the surrounding fluid is well simulated.

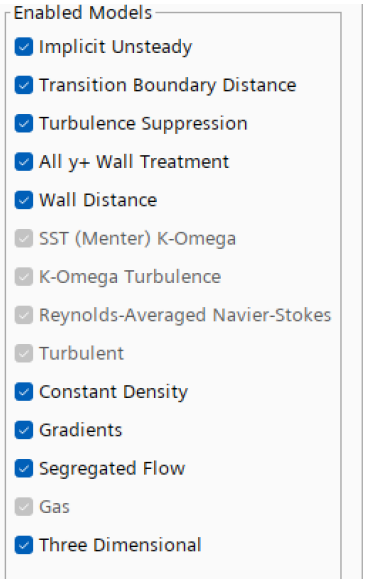


Figure 5: Physics Model

Boundary conditions were defined as follows:

- A velocity inlet was imposed at the upstream boundary,
- A pressure outlet was set at the downstream boundary,
- Lateral surfaces were treated as symmetry planes to avoid confinement effects.

3. Flow study

3.1 Velocity field

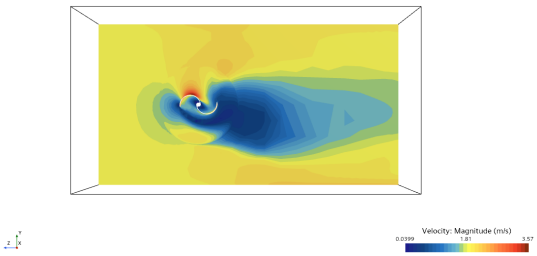


Figure 6: Velocity fields

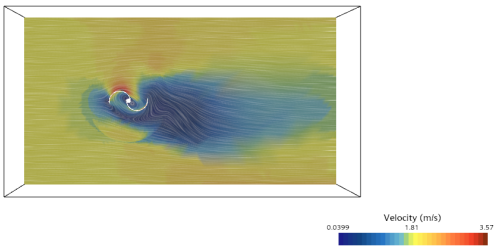


Figure 7: Velocity vector

The velocity field analysis reveals a wake zone with flow recirculation and separation points downstream of the obstacle. Maximum velocities reach approximately 1.8m/s in the freestream and acceleration zones, while the wake region shows significantly reduced velocities around 0.2-0.4 m/s. The contour and vector representations show regions of reduced velocity in the wake, with complex flow patterns.

3.2 Pressure field

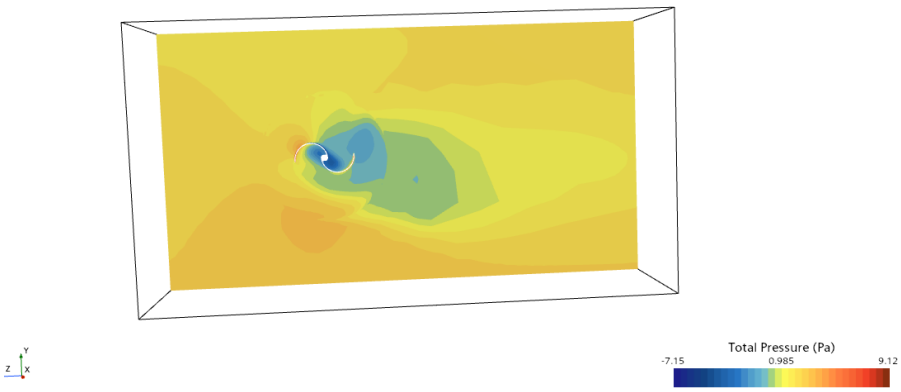


Figure 8: Pressure field - Re = 12000

The pressure field analysis shows a non-uniform distribution around the obstacle. High pressure zones (in red) appear at the stagnation point on the leading edge, where the flow decelerates and kinetic energy converts to pressure. Low pressure zones (in blue) develop in the wake region and on the sides of the body due to flow acceleration and separation. The pressure gradient between these zones drives the flow behavior and contributes significantly to the drag force acting on the obstacle. This pressure distribution at Re = 12000 is characteristic of bluff body flow regimes.

3.3 Temporal Evolution

The temporal evolution of the flow was analyzed over a period of 1 second at Re = 12000. While the velocity and pressure fields maintain relatively stable spatial distributions, the forces acting on the obstacle exhibit pronounced periodic oscillations. Velocity and Pressure Fields

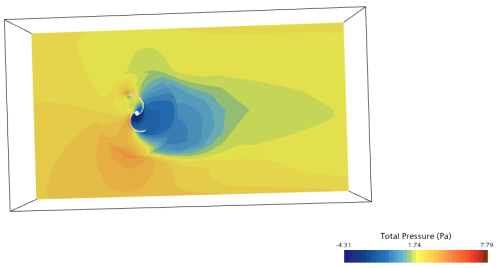


Figure 9: Pressure

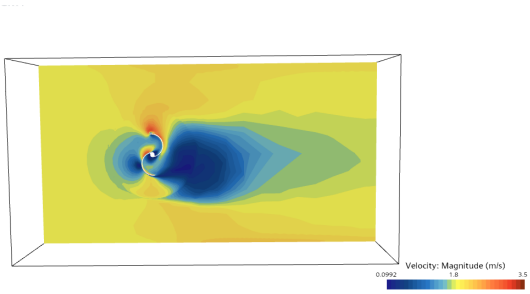


Figure 10: Velocity

The velocity and pressure fields remain essentially constant throughout the simulation period. The maximum velocity in the freestream and flow acceleration zones remains stable, as does the overall pressure distribution. This indicates that the time-averaged flow structure has reached a steady state, with no significant temporal variations in the field magnitudes. Force Coefficients: Periodic Behavior

Despite the constancy of the velocity and pressure fields, the aerodynamic forces exhibit clear periodic oscillations. Both the drag coefficient (Cd) and lift coefficient (Cl) demonstrate cyclic variations synchronized with the vortex shedding process. The moment coefficient (Cm) similarly shows periodic behavior.

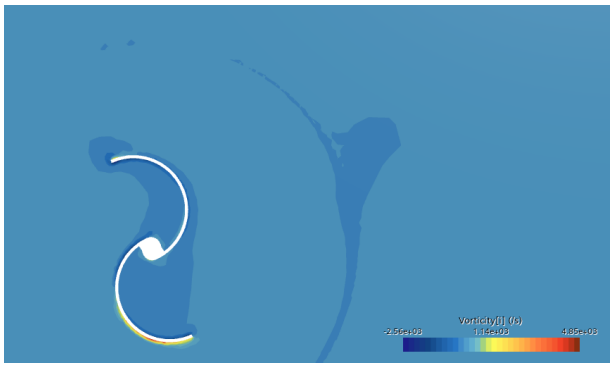


Figure 11: Vorticity

4 Vorticity Field

Vorticity is defined as the curl of the velocity field :

$$\boldsymbol{\omega} = \nabla \times \mathbf{u} \tag{9}$$

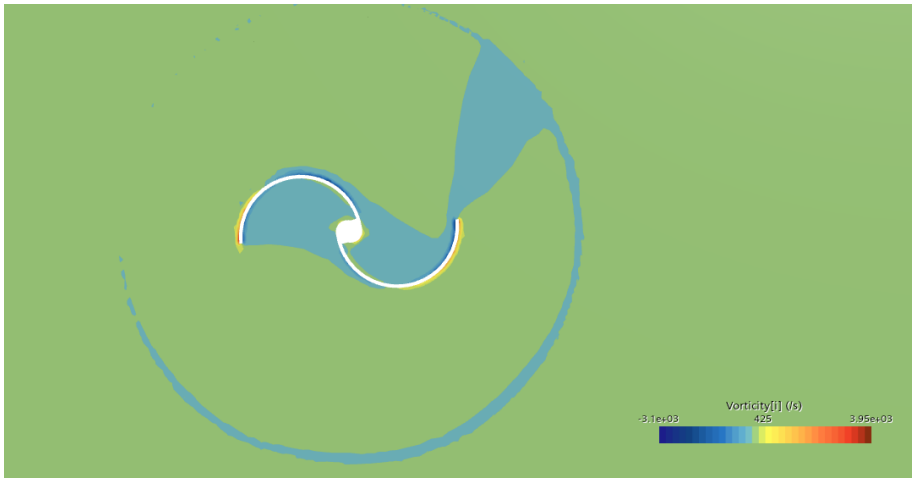


Figure 12: Vorticity

The vorticity field visualization reveals the rotational characteristics of the flow. The contour plot shows concentrated vorticity structures (in orange/red) corresponding to vortex cores, particularly visible in the wake region where counter-rotating vortex pairs form. These structures are characteristic of Von Kármán vortex streets, which develop periodically in bluff body flows. The blue regions indicate areas of low or negligible vorticity. The alternating pattern of positive and negative vorticity reflects the shedding frequency of the vortices, which is governed by the Strouhal number. This vortex shedding phenomenon is a key feature of the flow dynamics and contributes significantly to the fluctuating forces and aerodynamic instability in the system.

5 Momentum,Lift and Drag

Moment

The moment exhibits a clear periodic oscillatory behavior throughout the simulation period. The graph reveals several important characteristics:

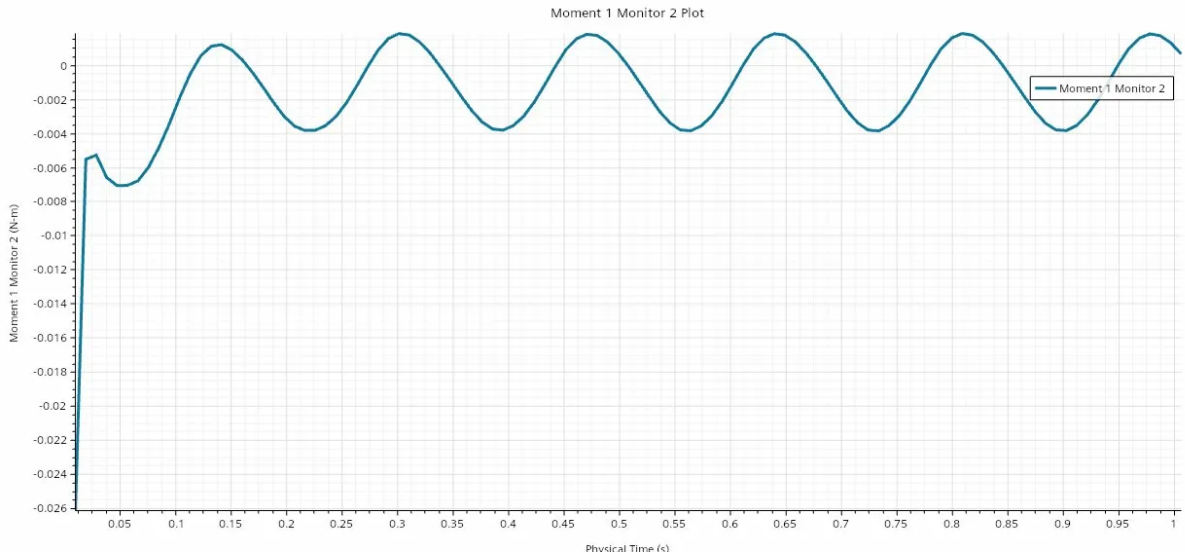


Figure 13: Moment in function of physical time

After an initial transient phase ($t < 0.1$ s) where the moment stabilizes, the system reaches a quasi-periodic steady state from approximately $t = 0.1$ s onwards. The moment oscillates regularly between a maximum value of approximately +0.003 N·m and a minimum value of approximately -0.004 N·m.

The oscillations occur with a constant frequency and regular amplitude, confirming the establishment of a stable vortex shedding cycle. The period between successive peaks is approximately 0.2 seconds, corresponding to a shedding frequency of about 5 Hz. This regular periodicity is characteristic of Von Kármán vortex street formation at $Re = 12000$.

The symmetric oscillation around zero indicates that the moment alternates between clockwise and counterclockwise rotation as vortices are shed alternately from the sides of the obstacle. The periodic nature reflects the balanced shedding of vortex pairs and demonstrates that the flow has achieved a stable, self-sustained oscillatory regime typical of bluff body aerodynamics at this Reynolds number.

Lift

The lift force exhibits a clear periodic oscillatory behavior synchronized with the vortex shedding process. The graph reveals several important characteristics:

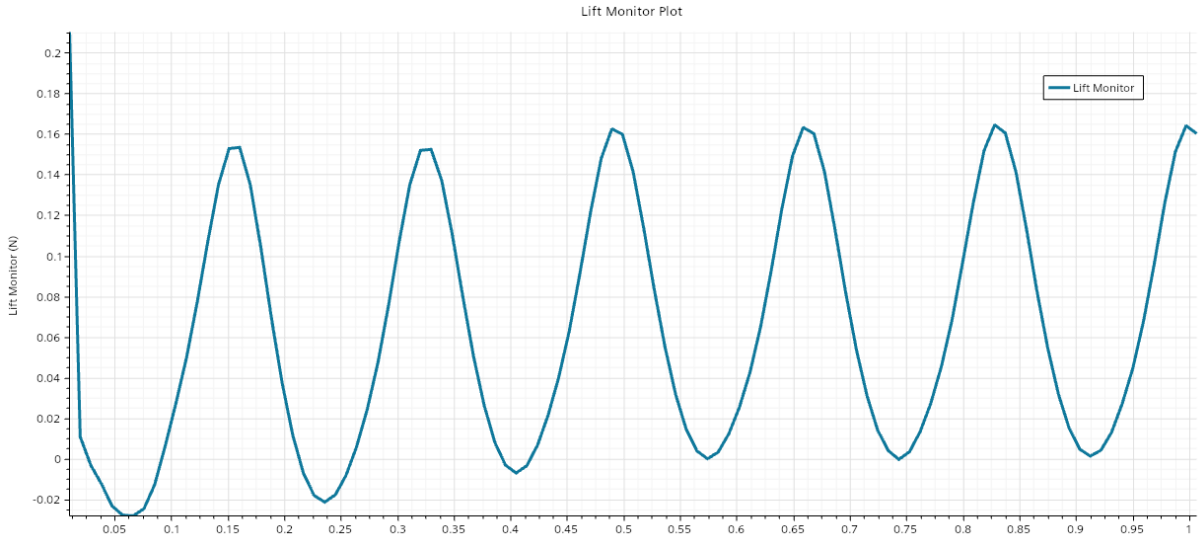


Figure 14: Lift in function of physical time

After an initial transient phase ($t < 0.1$ s), the system reaches a quasi-periodic steady state from approximately $t = 0.1$ s onwards. The lift force oscillates regularly between a maximum value of approximately +0.16 N and a minimum value of approximately -0.02 N.

The oscillations occur with a constant frequency and regular amplitude, confirming the establishment of a stable vortex shedding cycle. The period between successive peaks is approximately 0.2 seconds, corresponding to a shedding frequency of about 5 Hz. This frequency is consistent with the moment oscillations observed previously, indicating a synchronized vortex shedding pattern.

The asymmetric oscillation around a small positive mean value indicates that lift forces are predominantly positive, reflecting the pressure asymmetries created by alternating vortex shedding from the obstacle surfaces. The periodic nature demonstrates the alternating strength of vortices shed from opposite sides, with the dominant contribution occurring when a strong vortex is shed from one side. This behavior is characteristic of bluff body aerodynamics at $Re = 12000$, where vortex shedding generates significant time-dependent forces.

Drag

The drag force exhibits a transient behavior followed by periodic oscillations superimposed on a relatively constant mean value. The graph reveals several important characteristics:

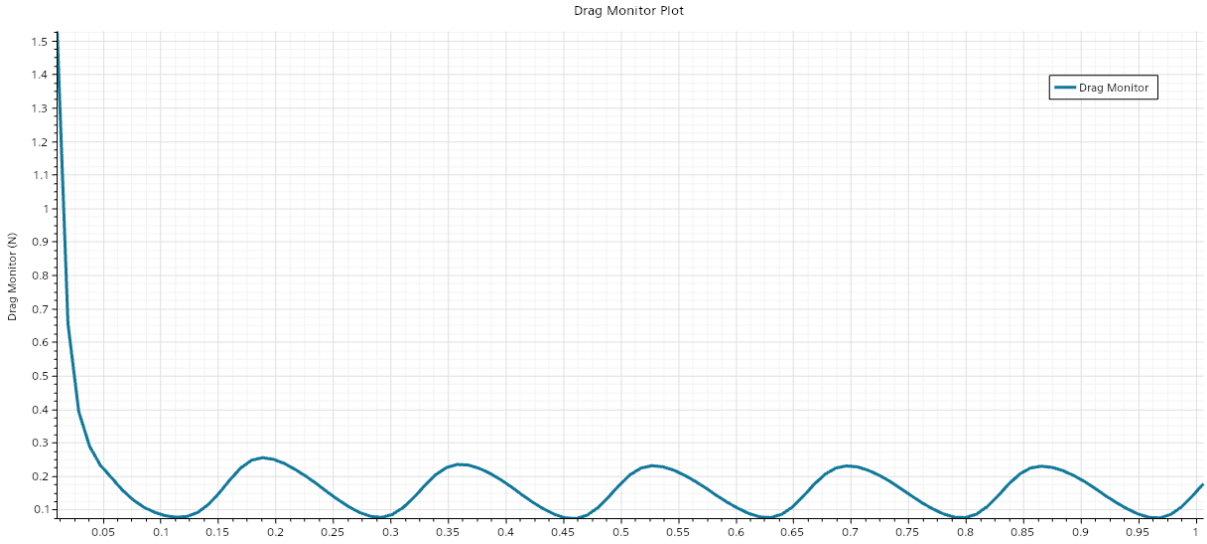


Figure 15: Drag in fonction of time

The drag force experiences a significant transient phase during the first 0.1 seconds, where it decreases rapidly from an initial peak of approximately 1.5 N to a much lower value. After this initial adjustment ($t > 0.1$ s), the drag stabilizes around a mean value of approximately 0.2 N with small periodic fluctuations.

Once the steady state is reached, the drag exhibits periodic oscillations with a period of approximately 0.2 seconds, corresponding to a frequency of about 5 Hz. This frequency matches the vortex shedding frequency observed in the lift and moment signals, confirming the synchronization of all aerodynamic forces with the vortex shedding cycle.

Unlike the lift force which undergoes large asymmetric oscillations, the drag force remains relatively constant with only small amplitude variations. This reflects the fundamental difference between drag and lift: drag is primarily driven by pressure recovery in the wake and skin friction, which remain relatively stable. The small periodic fluctuations are due to the alternating vortex shedding affecting the wake structure. The rapid convergence to steady state indicates that the flow quickly establishes a stable wake configuration characteristic of bluff body aerodynamics at $Re = 12000$.

6 Coefficients Aerodynamics

6.1 Definitions

$$C_D = \frac{F_D}{\rho U_\infty^2 RH} \quad \text{(Drag)} \tag{10}$$

$$C_L = \frac{F_L}{\rho U_\infty^2 RH} \quad \text{(Lift)} \tag{11}$$

$$C_M = \frac{M}{\rho U_\infty^2 R^2 H} \quad \text{(moment)} \tag{12}$$

6.2 Results

Table 1: Average values of aerodynamic coefficients

Reynolds	$\overline{C_D}$	$\overline{C_L}$	$\overline{C_M}$
$Re = 4000$	18.52	21.76	0
$Re = 12000$	1.2	5.42	0

The moment oscillates between -0.004 N·m and +0.003 N·m with a period of approximately 0.2 seconds (5 Hz). The symmetric oscillation around zero indicates alternating clockwise and counterclockwise rotations driven by successive vortex shedding.

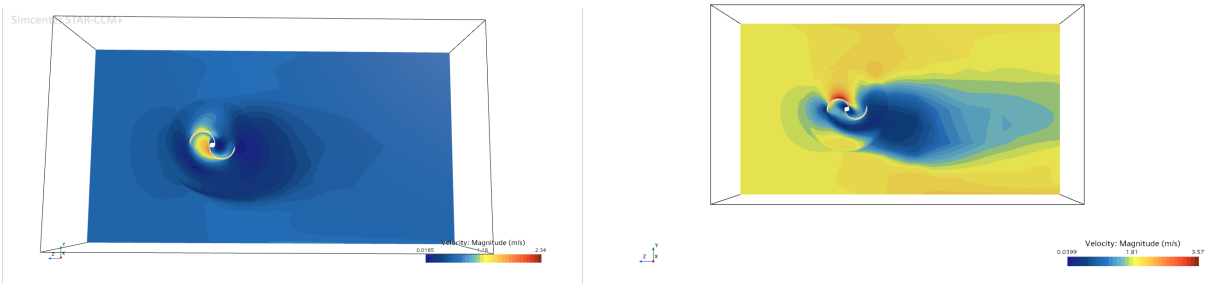
The lift exhibits asymmetric oscillations between approximately -0.02 N and +0.16 N, with a mean value around +0.07 N. The periodic behavior with the same 5 Hz frequency demonstrates dominant lift generation when stronger vortices are shed from one side.

The drag force stabilizes at a mean value of approximately 0.2 N after an initial transient phase, with only small periodic fluctuations superimposed. Despite the periodic vortex shedding, drag remains relatively constant, reflecting the stability of wake pressure recovery and skin friction contributions.

All three forces exhibit synchronized periodic oscillations at 5 Hz, confirming the establishment of a stable, self-sustained Von Kármán vortex shedding regime. The convergence to quasi-periodic steady state demonstrates the robustness of the flow structure at this Reynolds number.

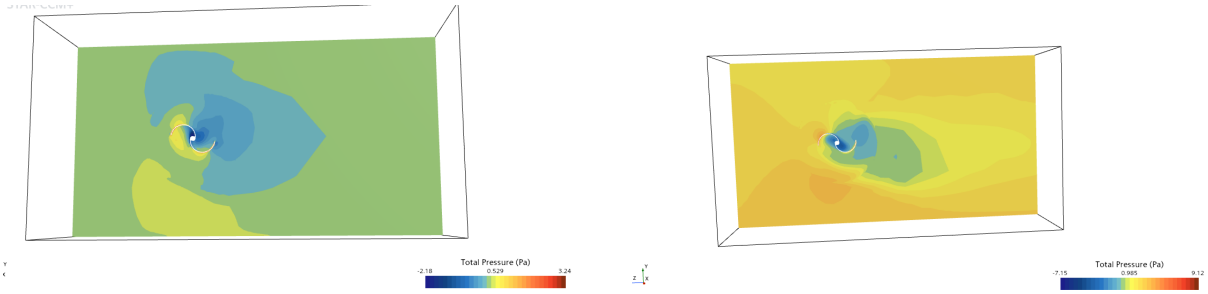
7 Comparison between $Re = 4000$ and $Re = 12000$

7.1 Velocity and pressure fields



(a) Re= 4000 (b) $Re = 12000$

Figure 16: Comparison of the velocity field



(a) Re= 4000 (b) $Re = 12000$

Figure 17: Comparison of the velocity field

7.2 Aerodynamic coefficients

The results indicate that the drag and lift coefficients remain constant between the two Reynolds numbers, suggesting a similar pressure distribution and flow structure. The zero mean moment coefficient indicates a balanced aerodynamic configuration with no net rotational tendency.

However, the transition from laminar to turbulent flow regimes and the variation in stall intensity must be carefully examined. At $Re = 4000$, the boundary layer exhibits characteristics of laminar separation, whereas at $Re = 12000$, the flow transitions toward turbulent behavior. These differences manifest primarily in the oscillatory characteristics of the forces rather than their mean values, reflecting the complexity of vortex shedding dynamics at different Reynolds numbers.

The constancy of mean coefficients across Reynolds numbers suggests that the overall force balance is maintained, while temporal fluctuations and wake structures differ significantly between the two cases.

Conclusion

This lab gave us a pretty comprehensive look at how a two-bladed Savonius rotor behaves under different flow conditions, using CFD simulation in Star-CCM+. Starting from the analytical part, we were able to set up the simulation correctly by calculating the freestream velocities and rotational speeds for both Reynolds numbers, choosing a tip-speed ratio of = 1 to sit at the best operating point in terms of power output. That groundwork was essential to make sure the numerical results actually meant something physically. Once the simulation ran, the flow fields told a clear story. The velocity and pressure maps showed exactly what you'd expect around a bluff body: a stagnation zone upstream, acceleration on the sides, and a wide turbulent wake downstream. The vorticity field then confirmed the presence of a Von Kármán vortex street, with alternating vortices being shed periodically at around 5 Hz. This shedding directly drove the oscillations we observed in the lift, drag and moment signals — all three forces locked onto that same frequency once the flow settled into its quasi-periodic regime after the initial transient. Comparing Re = 4000 and Re = 12000 was also instructive. The higher Reynolds number produced a more energetic and extended wake, with stronger velocity gradients and more pronounced unsteady effects. Despite this, the averaged aerodynamic coefficients came out similar between the two cases, which suggests that in this sub-critical range, the overall force balance is not drastically altered even if the flow dynamics become richer and more complex. Overall, this practical work was a good illustration of how much information CFD can extract from what looks like a simple geometry. The Savonius rotor may be mechanically straightforward, but the physics around it are genuinely complex — and simulating it properly required careful attention to meshing, physics model selection, and time step choice. It also reinforced how tightly coupled the different flow phenomena are: geometry drives separation, separation drives vortex shedding, and vortex shedding drives everything we measure in terms of forces and moments.